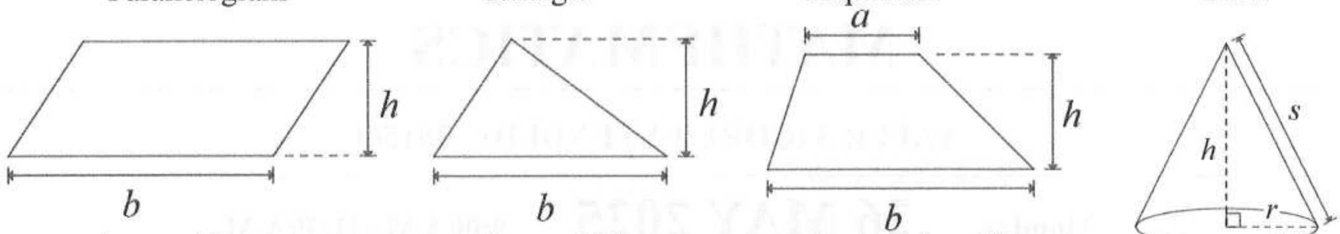
INFORMATION AND FORMULAE

MENSURATION

Parallelogram | Triangle | Trapezium | Cone

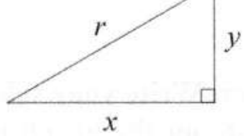


Area $= bh$ | Area $= \frac{1}{2}bh$ | Area $= \frac{1}{2}(a+b)h$

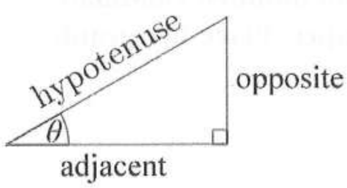
Circle (radius r, diameter d)	Circumference	$2\pi r$ or πd
	Area	πr^2
Cylinder (radius r, height h)	Volume	$\pi r^2 h$
	Area of curved surface	$2\pi rh$
Sphere (radius r)	Volume	$\frac{4}{3}\pi r^3$
	Area of curved surface	$4\pi r^2$
Prism	Volume	area of cross-section $\times$ length
Pyramid	Volume	$\frac{1}{3}\times$ area of base $\times$ height
Cone (radius r, height h)	Volume	$\frac{1}{3}\pi r^2 h$
	Area of curved surface	πrs where slant height $s = \sqrt{h^2 + r^2}$

TRIGONOMETRY

Right-angled Triangle:

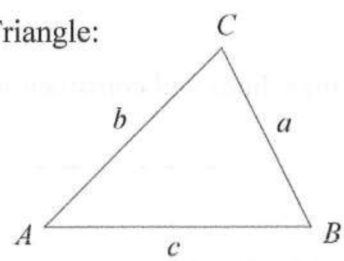


$r^2 = x^2 + y^2$ (Result of Pythagoras)



$\sin\theta = \dfrac{\text{opposite}}{\text{hypotenuse}}$, $\cos\theta = \dfrac{\text{adjacent}}{\text{hypotenuse}}$, $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}}$

Any Triangle:



In any triangle ABC: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$

$a^2 = b^2 + c^2 - 2bc\cos A$

$\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

Area of triangle $ABC = \frac{1}{2}ab\sin C$

NUMBER Standard Form is $a \times 10^n$ where $1 \le a < 10$ and n is an integer.

ALGEBRA The quadratic equation $ax^2 + bx + c = 0$ has solutions $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

The determinant of matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $ad - bc$.

The inverse of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $\dfrac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

If $y = ax^n$, then $\dfrac{dy}{dx} = anx^{n-1}$